THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 6: **Triangles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The diagonals of a rhombus are 10 and 24. Its side squared is: 1

(a) 169 **(b)** 168 **(c)** 171 **(d)** 170

2. Two poles of heights 20 m and 14 m are 8 m apart. The distance between their tops (m) is: 1

(a) 10 **(b)** 11 **(c)** 12 **(d)** 9

3. If $DE \parallel BC$ in $\triangle ABC$, $AB = 8$ cm, $AD = 2$ cm, $AC = 16$ cm, then EC (cm) is: 1

(a) 11 **(b)** 13 **(c)** 14 **(d)** 12

4. The areas of two similar triangles are 16 cm² and 81 cm². The ratio of their perimeters is 4: 1

(a) 10 **(b)** 9 **(c)** 11 **(d)** 8

5. The altitude to the hypotenuse of a right triangle divides it into 2 cm and 18 cm. The altitude (cm) is: 1

(a) 6 **(b)** 8 **(c)** 7 **(d)** 5

6. A pole 10 m high casts a shadow 6 m long. A tower casting a shadow 54 m has height (m): 1

(a) 90 **(b)** 89 **(c)** 92 **(d)** 91

7. The diagonals of a rhombus are 18 and 24. Its side squared is: 1

(a) 226 **(b)** 227 **(c)** 225 **(d)** 224

8. Two poles of heights 30 m and 6 m are 7 m apart. The distance between their tops (m) is: 1

(a) 25 **(b)** 26 **(c)** 27 **(d)** 24

9. If $DE \parallel BC$ in $\triangle ABC$, $AB = 9$ cm, $AD = 5$ cm, $AC = 18$ cm, then EC (cm) is: 1

(a) 7 **(b)** 9 **(c)** 10 **(d)** 8

10. The areas of two similar triangles are 50 cm² and 72 cm². The ratio of their perimeters is 5: 1

(a) 8 **(b)** 5 **(c)** 7 **(d)** 6

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Two similar triangles have perimeters 16 cm and 48 cm. If a side of the smaller is 5 cm, find the corresponding side of the larger and the ratio of their areas. 2

12. In $\triangle ABC$, D and E are the midpoints of AB and AC and $BC = 12$ cm. Find DE and the ratio of areas of $\triangle ADE$ and $\triangle ABC$. 2

13. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AB = 5$ cm, $AD = 2$ cm and $AC = 15$ cm, find AE and EC. 2

14. Two similar triangles have perimeters 10 cm and 40 cm. If a side of the smaller is 3 cm, find the corresponding side of the larger and the ratio of their areas. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AB = 9$ cm, $AD = 6$ cm and $AC = 18$ cm, find AE and EC. 3

16. $\triangle ABC \sim \triangle PQR$ with $AB = 4$ cm, $BC = 7$ cm, $CA = 8$ cm and $PQ = 8$ cm. Find QR and RP. 3

17. In a right triangle, the altitude to the hypotenuse divides it into segments of 3 cm and 27 cm. Find the altitude. 3

OR

In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AB = 13$ cm, $AD = 5$ cm and $AC = 26$ cm, find AE and EC. 3

18. $\triangle ABC \sim \triangle PQR$ with $AB = 6$ cm, $BC = 8$ cm, $CA = 10$ cm and $PQ = 24$ cm. Find QR and RP. 3

OR

In a right triangle, the altitude to the hypotenuse divides it into segments of 5 cm and 20 cm. Find the altitude. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two poles of heights 18 m and 6 m stand on level ground 5 m apart. Find the distance between their tops. 5

OR

The diagonals of a rhombus are 18 cm and 24 cm. Find its side. 5

20. Two poles of heights 25 m and 9 m stand on level ground 12 m apart. Find the distance between their tops. 5

OR

The diagonals of a rhombus are 16 cm and 30 cm. Find its side. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND